

The determinant

CHAPTER 6

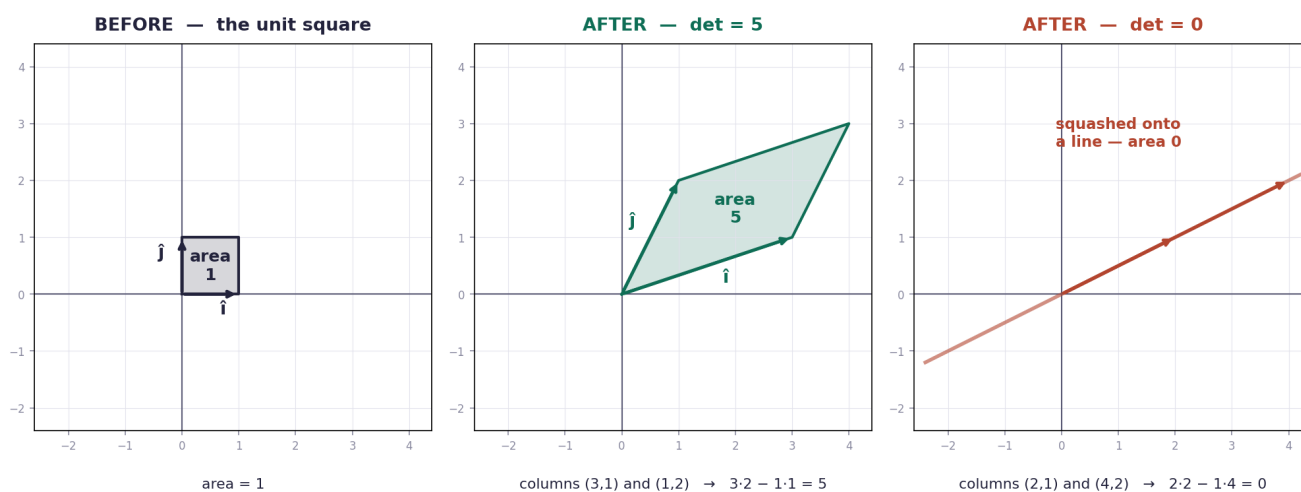
The determinant

One number that says how much space was stretched, squeezed, or destroyed.

THE IDEA

Start with the unit square formed by $\hat{i}$ and $\hat{j}$, area exactly 1. Apply a transformation and it becomes a parallelogram. Its new area is the **determinant** — and every region in the plane is scaled by that same factor, not just that square.

The determinant = how much the transformation scales AREA



READING THE ANSWER

Determinant	Meaning
det = 3	Areas triple
det = 1	Area unchanged (e.g. a pure rotation)
det = 0.5	Areas halve — space squeezed
det = 0	Space collapses onto a line or point — information destroyed
det negative	Space is flipped over ; the size is $ \text{det} $

THE FORMULA

For a matrix with columns (a, c) and (b, d): **det = ad - bc**

WORKED EXAMPLES

CHAPTER 6 — THE DETERMINANT

$\det$ = how much the transformation scales AREA

THE FORMULA

for $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$: **$\det = a \cdot d - b \cdot c$**

a, c = first column b, d = second column

EXAMPLE 1

columns (3, 1) and (1, 2) $\rightarrow \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$

$$a = 3 \quad b = 1 \quad c = 1 \quad d = 2$$

$$\det = (3)(2) - (1)(1)$$

$$= 6 - 1$$

$$= 5$$

areas become 5 times bigger

EXAMPLE 2 (the important case)

columns (2, 1) and (4, 2) $\rightarrow \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$

$$a = 2 \quad b = 4 \quad c = 1 \quad d = 2$$

$$\det = (2)(2) - (4)(1)$$

$$= 4 - 4$$

$$= 0$$

$$(4, 2) = 2 \times (2, 1) \rightarrow \text{columns are DEPENDENT}$$

space collapses onto a line $\rightarrow$ no inverse exists

WHAT THE ANSWER MEANS

$\det = 3$ areas triple

$\det = 1$ area unchanged (e.g. rotation)

$\det = 0.5$ areas halve — squeezed

$\det = 0$ collapsed to a line — NO inverse

$\det = -2$ flipped over, areas doubled

CAREFUL: $ad - bc$ is products THEN subtraction.

Write a, b, c, d labelled first — then compute.

The link that matters: $\det = 0$ means the columns are **linearly dependent** (Chapter 2), space collapses to a lower dimension, two different inputs can land on the same output — so there is **no way back**, and the matrix has **no inverse**. That is the doorway to Chapter 7.

Careful: $ad - bc$ is two products *then* a subtraction. Write a, b, c, d labelled first, then compute.

FORMULAS FROM THIS CHAPTER

Determinant (2×2)

$$\det = ad - bc$$

Meaning	factor by which areas are scaled
det = 0	columns dependent · space collapses · no inverse
det negative	space flipped over; size is det